BUSINESS INTELLIGENCE AND DATA ANALYTICS
– WE PREDICT THE FUTURE –

University of Petra

307401
Big Data and Data Warehouses

Introduction to Hadoop

Types of Processing in Computing

Processing Type	Description	Example
Sequential Processing	Tasks are executed one after another in a linear sequence.	Reading and executing lines of code in a single-threaded program.
Multi-Tasking	Multiple tasks share the same CPU by switching between them rapidly.	Running multiple applications (browser, music player, and document editor) on a computer.
Parallel Processing	Multiple processors or cores execute different parts of a task simultaneously.	Image processing using multiple CPU cores.
Distributed Processing	Tasks are divided across multiple computers working together over a network.	Cloud computing and blockchain networks.

Speedup Concept

Formula:

$$Speedup(S) = \frac{T_1}{T_p}$$

Where:

- T_1 = Execution time with a single processor.
- T_p = Execution time with p processors.

Key Points:

- **Definition:** Speedup measures the performance gain achieved by using multiple processors compared to a single one.
- **Linear Speedup:** Ideal scenario where $S = p$ (e.g., 4 processors give 4× speedup).
- **Superlinear Speedup:** Occasionally, $S > p$ due to better memory utilization or caching effects.
- **Sublinear Speedup:** Speedup is less than the number of processors due to communication and synchronization overhead.
- **Amdahl's Law:** The speedup is limited by the sequential portion of the program:

$$S = \frac{1}{(1-P) + \frac{P}{p}}$$

- **Gustafson's Law:** Speedup increases with problem size, making large-scale distributed computing more efficient:

$$S = p - (1-P) \times (p-1)$$

Introduction to Hadoop

- Hadoop is an open-source framework overseen by Apache Software Foundation which is written in Java for storing and processing of huge datasets with the cluster of commodity hardware.
- There are mainly two problems with the big data that Hadoop addresses:
  1. The need to store huge amount of data.
  2. The need to process that stored data.
- The traditional approach like RDBMS is not sufficient due to the heterogeneity of the data.
- Hadoop serve as the solution to the problem of big data i.e. storing and processing big data.

History of Hadoop

Doug Cutting

Mike Cafarella

- Early Beginnings (2002-2003):
  - Nutch Project: Open-source search engine project by Doug Cutting and Mike Cafarella.
  - Challenges in managing and processing large volumes of web data.
- Google's Influence (2003-2004):
  - Google Papers: Introduction of Google's GFS (Google File System) and MapReduce.
  - Inspiration for Hadoop's core concepts.
- Birth of Hadoop (2005-2006):
  - Creation of Hadoop as a separate project within Apache Nutch.
  - Named after Doug Cutting's son's toy elephant.
- Yahoo! began investing in and contributing to Hadoop's development.
- Apache Hadoop (2008):
  - Hadoop became an Apache top-level project.
  - Core components: HDFS (Hadoop Distributed File System) and MapReduce.
  - Rapid growth in community contributions and adoption.
- Ecosystem Expansion (2009-2011):
  - New projects: Hive, Pig, HBase, Zookeeper.
  - Formation of companies like Cloudera for commercial support.
- Industry Adoption (2012-2014):
  - Mainstream use by major companies: Facebook, LinkedIn, eBay.
  - Introduction of Hadoop 2.0 with YARN (Yet Another Resource Negotiator).
  - Hadoop Summit: Annual conference for advancements and use cases.
- Modern Era (2015-Present):
  - Widespread enterprise adoption for various Big Data applications.
  - Integration with cloud platforms: AWS EMR, Google Cloud Dataproc.
  - Ongoing development and enhancements to the Hadoop ecosystem.
- Impact of Hadoop:
  - Pioneered the Big Data revolution.
  - Enabled scalable, distributed data processing.
  - Fostered a strong open-source community and ecosystem.

Hadoop Echo System

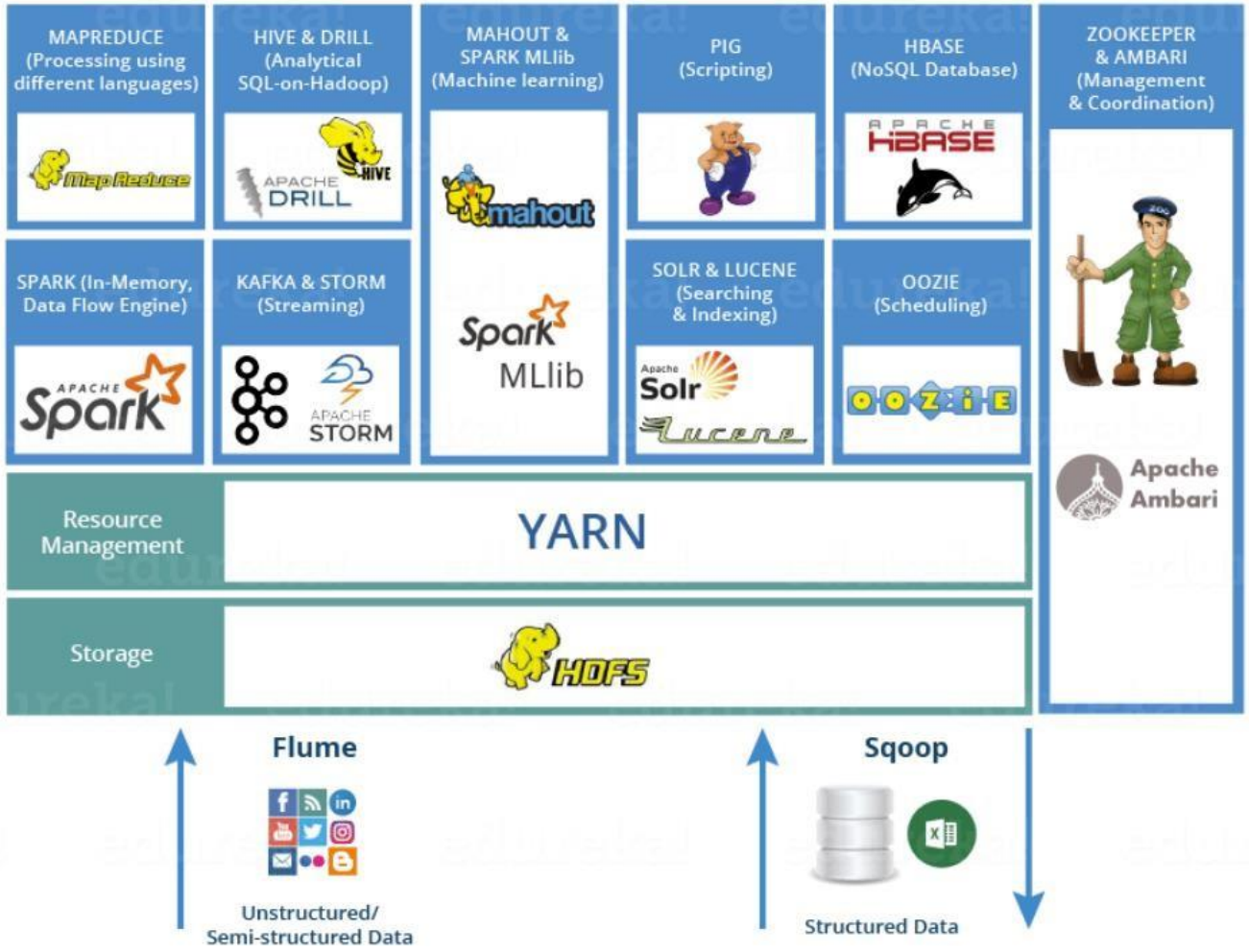
- Hadoop ecosystem is a suite of tools and frameworks

that enables Hadoop to handle large amounts of data.

Basically, Hadoop has the following three components:

1. HDFS (Hadoop Distributed File System): A distributed file system that provides high-throughput access to application data.
2. MapReduce: A framework for parallel processing of large data sets.
3. YARN, Resource Manager.

Other ecosystem components: Hive, Pig, HBase, Storm, Spark, Kafka, Flume, Sqoop, Oozie, Zookeeper…etc. These components **work together** to provide a comprehensive solution for managing and processing large data sets in a distributed computing environment.



HDFS – Hadoop Distributed File System

What is HDFS?

- **Hadoop Distributed File System (HDFS)** is the storage layer in Hadoop.
- It enables **distributed storage** and **fault tolerance** by replicating data across multiple nodes.

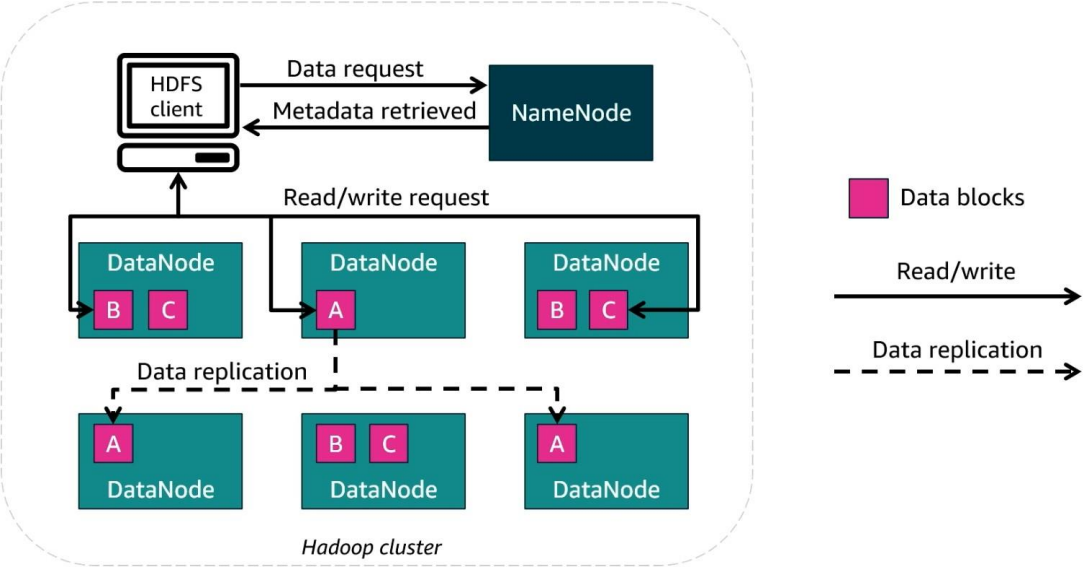
- **Key Components of HDFS**

1. NameNode

- Acts as the **master node**.
- Manages **metadata** (file locations, permissions, and structure).
- Does not store actual data.

2. DataNodes

- Store actual data in **blocks**.
- Handle read/write requests from clients.
- Perform **data replication** to ensure fault tolerance.



How HDFS Works?

1. **Client requests data** from HDFS.
2. **NameNode retrieves metadata** and provides block locations.
3. **Client reads/writes data blocks** directly from/to the DataNodes.
4. **Data replication** occurs across multiple DataNodes for redundancy.

YARN, Yet Another Resource Negotiator

What is YARN?

- YARN (Yet Another Resource Negotiator) is the resource management layer in Hadoop.
- It efficiently allocates resources and schedules tasks for processing big data jobs.

Key Components of YARN

1. Resource Manager (RM) - Central component managing resource allocation.

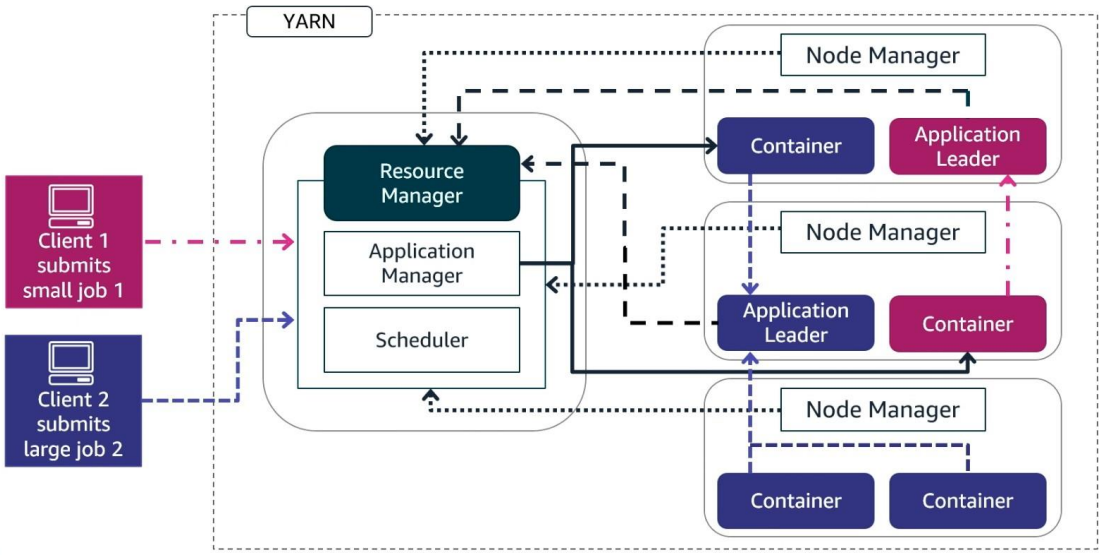
- It has two main parts:
  - **Application Manager** – Manages job submission and execution.
  - **Scheduler** – Allocates resources to tasks based on policies.

2. Node Managers (NM)

- Runs on each DataNode to execute tasks.
- Manages **Containers**, which are allocated resources for executing tasks.

3. Application Master (AM)

- Runs within a container and coordinates job execution.
- Communicates with the Resource Manager and Node Managers.

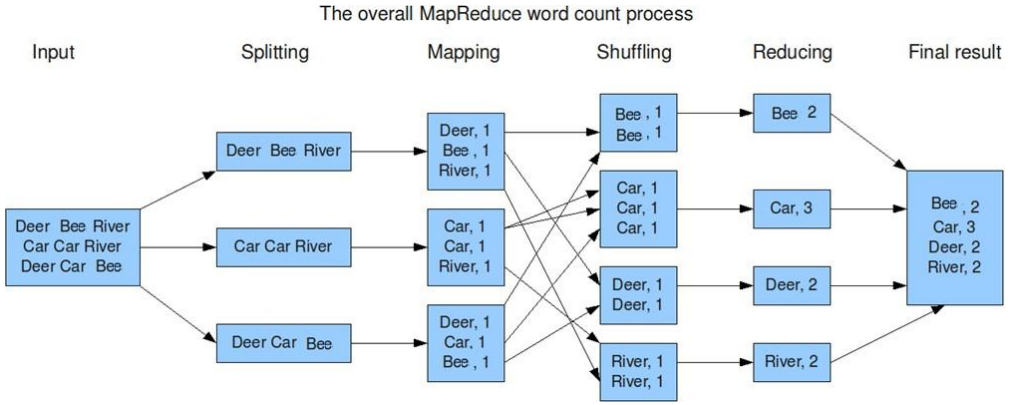


How YARN Works?

1. **Clients submit jobs** to the Resource Manager.
2. **Scheduler assigns resources** based on job size and priority.
3. **Application Master is launched** in a container to oversee execution.
4. **Node Managers execute tasks** in their respective containers.
5. **Results are sent back to clients** after completion.

MapReduce

- Map Reduce is a programming model for processing large data
- sets with a parallel, distributed algorithm on a cluster.
- Key Components:
  - Mapper: Takes a set of data and converts it into another set of data, where individual elements are broken down into key/value pairs.
  - Reducer: Takes the output from a mapper as an input and combines those data tuples into a smaller set of tuples.
- Workflow:
  - Input Splits: The input file is split into multiple pieces.
  - Map Phase: Processes the input splits and produces intermediate key-value pairs.
  - Shuffle and Sort Phase: Organizes the intermediate data by key.
  - Reduce Phase: Aggregates the sorted key-value pairs and generates the final output.



Real-World Use Case: Log Analysis for Ad Click Fraud Detection

Company Example: Facebook
Objective: **Detect fraudulent ad clicks** by analyzing massive volumes of server logs.
Data Source: Web server logs from Facebook's ad delivery infrastructure.

Data Structure (Sample Log Entry):


```
{  "timestamp": "2025-03-20T14:33:21Z",
  "user_id": "u98374",
  "ad_id": "ad4562",
  "ip_address": "192.168.1.5",
  "click": true,
  "device_type": "mobile",
  "location": "New York, USA"}
```


Processing with MapReduce:
Map Phase:
- Extract relevant fields: user_id, ip_address, ad_id, timestamp.
- Emit key-value pairs like (ip_address, 1) to count clicks per IP.

Reduce Phase:
- Aggregate counts of clicks per IP address.
- Identify IPs with suspiciously high click counts in a short time frame.

Outcome:
- Detects click patterns that suggest automated scripts or bot activity.
- Enables the system to block fraudulent IPs and protect ad revenue.

Apache Hive

What is Hive?

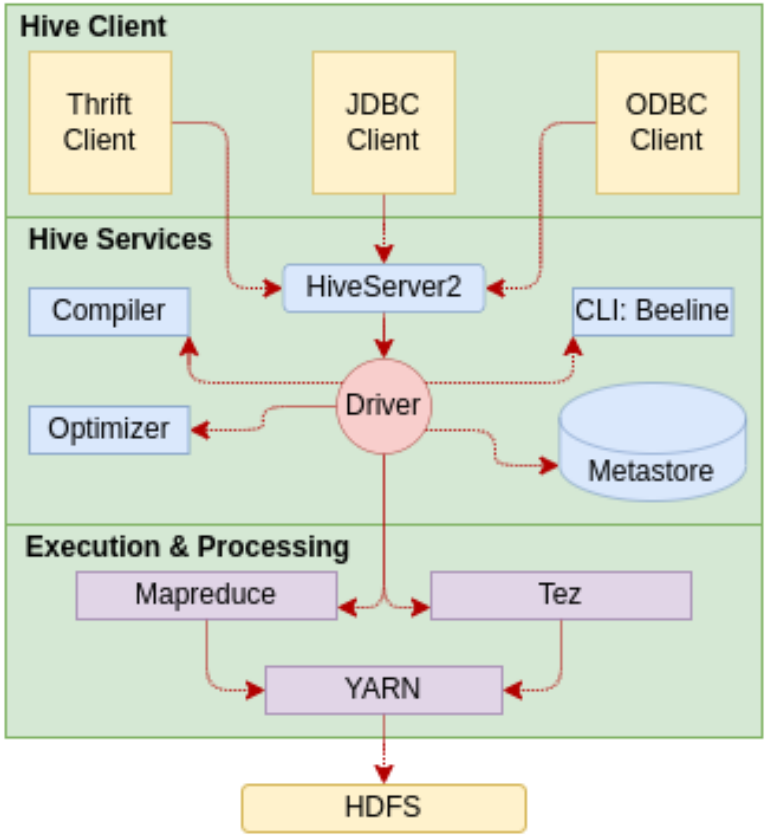
- Hive is a data warehouse system built on top of Hadoop.
- It allows querying and managing large datasets using a SQL-like language called HiveQL.
- Created by Facebook to simplify complex MapReduce programming for analysts.

Why Hive?

- Writing raw MapReduce code in Java is complex and time-consuming.
- Hive allows users to write declarative queries in HiveQL.
- Hive compiles these queries into efficient MapReduce tasks.

Hive Architecture

- Metastore: Stores metadata about Hive tables (schemas, locations, etc.).
- Driver: Manages the lifecycle of a HiveQL query and its execution plan.
- Compiler: Translates HiveQL into MapReduce jobs to run on Hadoop.



Real-World Use Case: Marketing Campaign Analysis

Company Example: NETFLIX

Objective: **Analyze customer behavior** to evaluate marketing campaign effectiveness.

Data Source: User activity logs stored in HDFS, including views, clicks, watch duration, and subscription status.

Data Structure (Hive Table Schema):

```
CREATE TABLE user_activity (
  user_id STRING,
  timestamp TIMESTAMP,
  action STRING,        -- e.g., "play", "pause", "subscribe",
"click_ad"
  content_id STRING,
  device_type STRING,
  campaign_id STRING)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```

Processing with Hive (HiveQL):
Perform aggregations:

- Total views per campaign
- Conversion rates from ad click to subscription
- Device-specific campaign performance

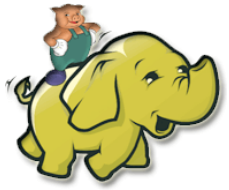
Sample Query:

```
SELECT campaign_id, COUNT(*) AS views, SUM(CASE WHEN action
= 'subscribe' THEN 1 ELSE 0 END) AS subscriptions FROM
user_activity WHERE action IN ('click_ad', 'subscribe') GROUP BY
campaign_id;
```

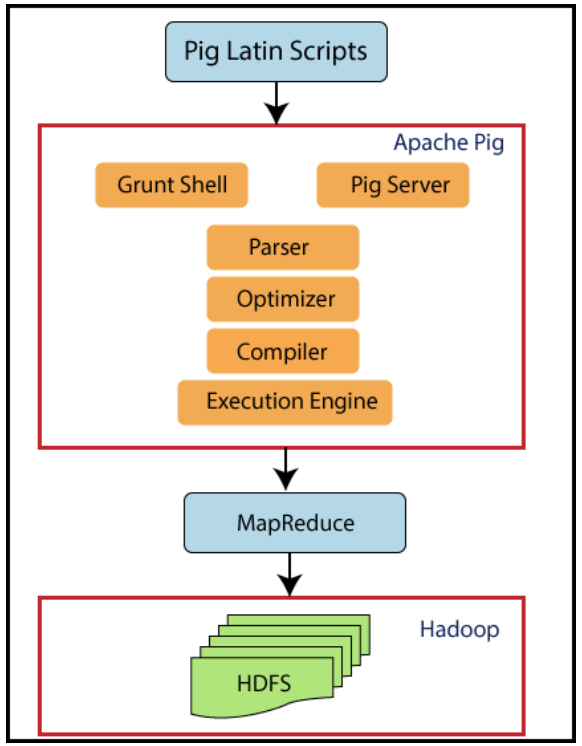
Outcome:

- Identifies high-performing campaigns and channels.
- Supports data-driven decisions for targeting and ad spend allocation.
- Let me know if you'd like this turned into a presentation slide or want a Hive use case from retail, telecom, or IoT.

Apache PIG – High Level SQL Like Language on top of Hadoop



- Run on Hadoop and translates Pig Latin scripts into MapReduce jobs.
- Created by YAHOO! to solve MapReduce challenges
- Pig Latin is a high-level data processing language that simplifies writing complex MapReduce code directly, this allows users to work with big data using simpler syntax compared to Map Reduce.
- Used for data transformations and analysis tasks over large datasets in Hadoop.
- **Architecture**:
  - **Pig Latin Scripts**: Define data transformation tasks.
  - **Pig Engine**: Executes these scripts on Hadoop.



https://www.analyticsvidhya.com/blog/2022/06/getting-started-with-apache-pig/

Hive vs Pig

Real-World Use Case: Web Log Processing for User Behavior Analysis

Company Example: *LinkedIn (used Pig before transitioning to Spark)*
Objective: Analyze web server logs to understand user navigation paths and session behavior.

Data Source:

- Daily web server log files stored in HDFS.
- Data Structure (Sample Log Fields):
- timestamp, user_id, session_id, page_url, referrer_url, device_type

Example Data:
2025-03-22T12:01:45Z, u12345, s7890, /jobs, /home, mobile

Processing with Apache Pig:
Load and Transform Logs:

```
logs = LOAD '/logs/2025/03/22'
    USING PigStorage(',')
    AS (timestamp:chararray, user_id:chararray,
session_id:chararray,
        page_url:chararray, referrer_url:chararray,
device_type:chararray);
```

Group and Analyze Sessions:

```
grouped = GROUP logs BY session_id;
session_counts = FOREACH grouped GENERATE group AS
session_id,
            COUNT(logs) AS page_views;
```

Filter High-Engagement Sessions:

```
engaged_sessions = FILTER session_counts BY page_views > 5;
```

Outcome:

- Identifies highly engaged users and frequent navigation paths.
- Insights used for UX improvements and targeted content placement.

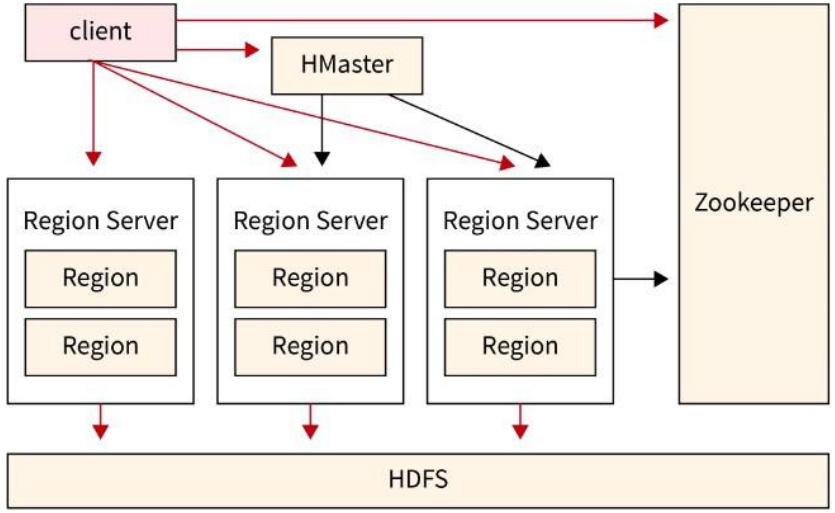
Apache HBase - Distributed Database on top of Hadoop

What is HBase?

- HBase is a distributed, column-oriented NoSQL database built on top of Hadoop HDFS.
- Designed for sparse, semi-structured, and unstructured data.
- Provides real-time read/write access to large datasets.
- Inspired by Google's Bigtable.

Key Concepts

- Row Key: Unique identifier for each row (like a primary key).
- Column Family: Logical group of columns stored together on disk.
- Column Qualifier: Actual field name inside a column family.
- Cell: Combination of Row Key + Column Family + Qualifier + Timestamp → Value.
- Timestamp: Each cell value is versioned by time.



Apache HBase Architecture

Architecture:

- **HMaster**: Manages the cluster.
- **RegionServer**: Handles read/write requests for regions of the database.
- **ZooKeeper**: Manages distributed coordination.

https://www.scaler.com/topics/hadoop/hbase-architecture/

Insert Data in HBase

- In HBase, we create tables and define column families.
- We must comply with column families when we insert data into the table.
- Column families are employed for storage and fast retrieval – HBase is Column Oriented Database.
- But we can use different column names within the column family e.g.

put 'employees', 'emp1', 'personal:name', 'Alice'	**emp1 has only name in personal**
put 'employees', 'emp2', 'personal:phone', '123456789'	emp2 has only phone in personal
put 'employees', 'emp3', 'job:salary', '60000'	emp3 has only salary in job

✅ **Schema-less?** Yes, **for columns** (qualifiers) – flexible and dynamic.

🧱 **Schema-fixed?** Yes, **for column families** – must be defined when the table is created.

🗂️ **Table Name:** `employees`

📦 **Column Families:** `personal`, `job`

✅ **1. Create the Table**

bash Copy Edit

```bash
create 'employees', 'personal', 'job'
```

✅ **2. Insert Data for Multiple Employees**

bash Copy Edit

```bash
# Employee 1: Alice
put 'employees', 'emp1', 'personal:name', 'Alice Smith'
put 'employees', 'emp1', 'personal:email', 'alice@example.com'
put 'employees', 'emp1', 'job:department', 'Engineering'
put 'employees', 'emp1', 'job:salary', '72000'

# Employee 2: Bob
put 'employees', 'emp2', 'personal:name', 'Bob Johnson'
put 'employees', 'emp2', 'personal:email', 'bob@example.com'
put 'employees', 'emp2', 'job:department', 'Marketing'
put 'employees', 'emp2', 'job:salary', '65000'

# Employee 3: Clara
put 'employees', 'emp3', 'personal:name', 'Clara Lee'
put 'employees', 'emp3', 'personal:email', 'clara@example.com'
put 'employees', 'emp3', 'job:department', 'Finance'
put 'employees', 'emp3', 'job:salary', '71000'
```

HBase Internal Row Data Shape

Each entry in HBase is stored as a **triplet (technically a quintuple):**

```pgsql
(Row Key, Column Family, Column Qualifier, Timestamp) => Value
```

This means **each field you insert becomes a cell**, and there's no fixed schema for columns like in a relational table.

Example: Employee Data Internally in HBase

Let's say you added this:

```bash
put 'employees', 'emp1', 'info:name', 'Alice Smith'
put 'employees', 'emp1', 'info:salary', '72000'
```

Internally, it looks more like:

Row Key	Column Family	Column	Timestamp	Value
emp1	info	name	1679553600000	Alice Smith
emp1	info	salary	1679553600000	72000

> HBase auto-generates timestamps unless you set them manually.

Real-World Use Case: Messaging Platform – Real-Time Message Storage & Retrieval

Company Example: *WhatsApp*

Objective: Store and retrieve billions of real-time chat messages with low latency and high throughput.

Data Source: User chat messages from mobile devices, synced to a backend service.

Data Structure (HBase Table Design):
Table: messages
Row Key: user_id + timestamp
Column Families:
- **meta: sender_id, receiver_id, message_type**
- **content: message_text, attachments**
- **status: delivered, read, timestamp**

Sample Record (in HBase):
Row Key: user123|2025-03-22T14:33:21Z
Column Family: meta
- sender_id: user123
- receiver_id: user456
Column Family: content
- message_text: "Hey, are we still meeting today?"
Column Family: status
- delivered: true
- read: false

Processing with HBase:
- Messages are written and read with low latency.
- Efficient for real-time features: chat history retrieval, unread message counts.
- Scales horizontally to billions of rows (messages) per day.

Outcome:
- Ensures seamless chat experience across devices.
- Supports real-time sync, delivery status tracking, and storage durability.

HBase vs Relational DB

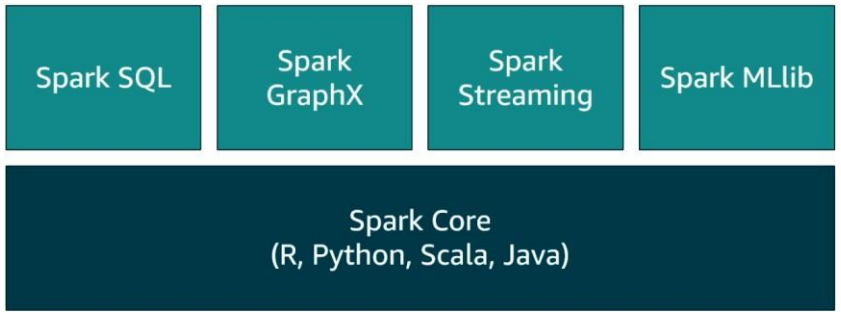
Feature	HBase	Relational DB
Schema	Flexible (schema-less columns)	Fixed schema
Row Structure	Rows can have different columns	Rows have same set of columns
Access Pattern	Real-time random read/write	Typically, transactional
Query Language	API (Java, REST, Thrift)	SQL
Storage Model	Column-oriented, sparse	Row-oriented
Best Use Case	Time-series, logs, IoT, real-time	Structured, normalized data

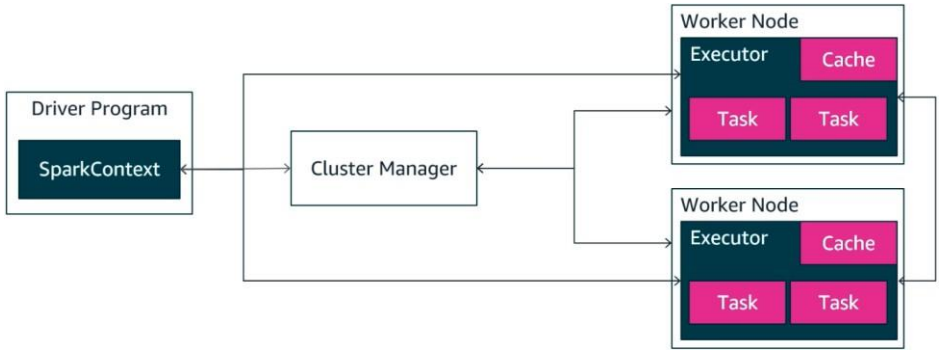
Apache Spark

- Apache Spark is an open-source distributed data processing engine optimized for large-scale data processing.
- In-Memory Processing: Spark keeps data in memory between transformations, speeding up processing times significantly compared to disk-based MapReduce.
- Spark can handle both batch and real-time processing.
- Spark offers APIs for various languages (Java, Scala, Python, R).

Core Components:

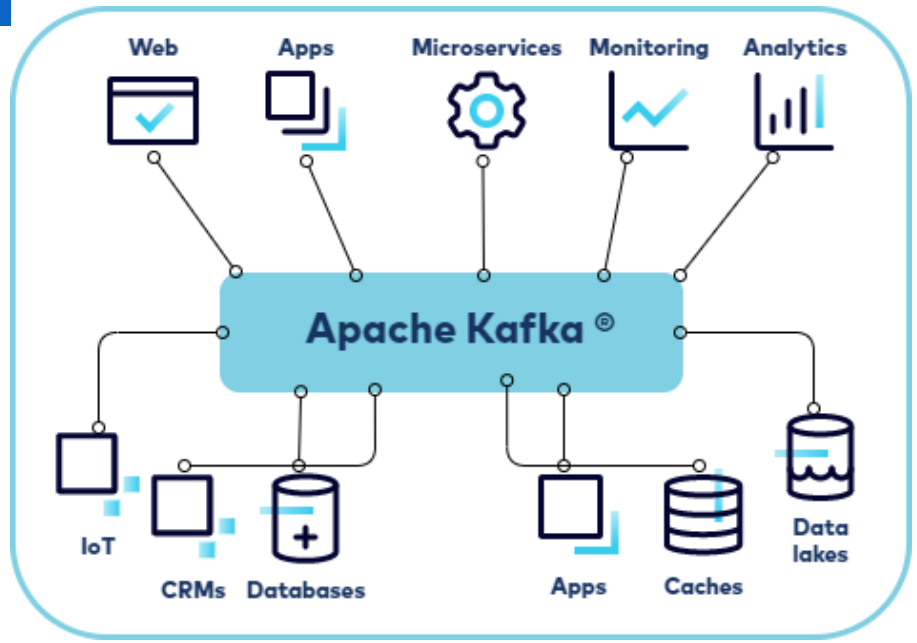
1. **Spark Core:** The foundation for Spark's API, handling basic data processing and distributing tasks across the cluster.
2. **Spark SQL:** Enables SQL-like queries on data stored in distributed datasets (DataFrames and Datasets).
3. **Spark Streaming:** Allows processing of real-time data streams, enabling near real-time analytics.
4. **MLlib:** Spark's machine learning library, offering tools for scalable machine learning.
5. **GraphX:** A graph processing library for large-scale graph computations.





Apache Kafka – Messaging System

- Apache Kafka is a distributed event streaming platform designed to handle high-throughput, real-time data feeds.
- Originally developed by LinkedIn, Kafka has become the backbone for data pipelines and real-time analytics in many large-scale systems.
- It serves as a message broker that allows various applications to produce and consume streams of data reliably, efficiently, with low latency and at scale.
- **Architecture**:
  - **Producer:** An application that sends data to Kafka topics.
  - **Consumer:** An application that reads data from Kafka topics.
  - **Broker:** A Kafka server that stores data and serves it to consumers.
  - Kafka clusters consist of multiple brokers.
  - **Topic:** A category or feed name to which records are published. Each topic can have multiple partitions, helping Kafka to parallelize data across the cluster.



Real-World Use Case: Real-Time Fraud Detection in Banking

Company Example: *Bank of America*

BANK OF AMERICA

Objective:
Detect fraudulent credit card transactions in real time using streaming data.

Data Source:
Credit card transaction stream from multiple sources (POS terminals, ATMs, mobile apps)

Data Structure (Sample Kafka Message):


```
{  "transaction_id": "txn78923",
  "timestamp": "2025-03-20T14:33:21Z",
  "card_id": "card3746",
  "amount": 324.55,
  "merchant": "Amazon",
  "location": "San Francisco, USA",
  "channel": "e-commerce"}
```


Processing with Kafka:

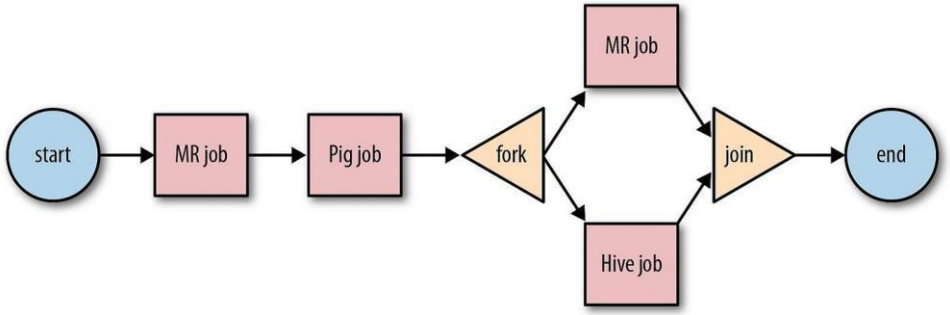
Producers:
POS terminals and mobile apps publish transaction data to Kafka topics (transactions, locations, users).

Kafka Streams / Consumers:
- Stream joins with historical user behavior data.
- Apply real-time rule-based or ML-based fraud detection (e.g., unusual amount, location mismatch, rapid transactions).
- Flag anomalies and push alerts to a fraud_alerts topic.

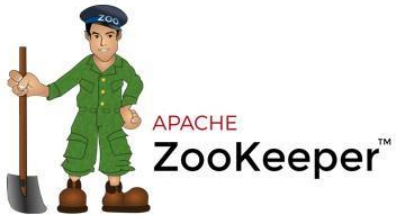
Outcome:
- Enables real-time detection and blocking of suspicious transactions.
- Reduces fraud loss and enhances customer trust.

Apache Oozie

- Apache Oozie is a workflow scheduler system designed to manage and coordinate Hadoop jobs in a sequence or directed acyclic graph (DAG) for end-to-end workflow management.
- Oozie is particularly useful for automating complex data workflows, managing dependencies, and orchestrating multiple jobs across the Hadoop ecosystem (such as MapReduce, Hive, Pig, and Sqoop jobs).
- It supports both time-based (frequency) and data-based triggers,
- making it versatile for different types of big data applications.
- **Main Functions:**
  - **Workflow Management**: Defines and manages job sequences.
  - **Coordination**: Manages job dependencies and schedules based on time and data availability.
  - **Retry Mechanism**: Automatically retries failed jobs.
- **Use Cases**: ETL workflows, batch processing, and managing complex job dependencies.

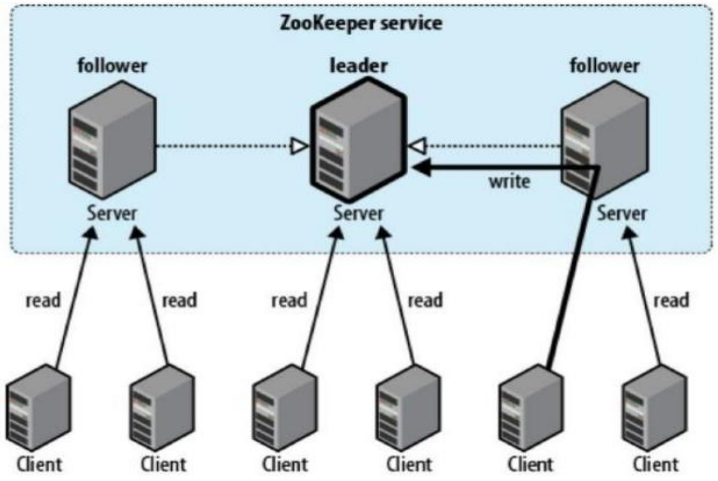


Apache Zookeeper



- Apache ZooKeeper is a centralized service designed for coordinating distributed applications.
- ZooKeeper plays a critical role in the Hadoop ecosystem and other distributed systems by helping to maintain the distributed state and providing a reliable way for services to work together.
- **Key Features of Apache ZooKeeper**

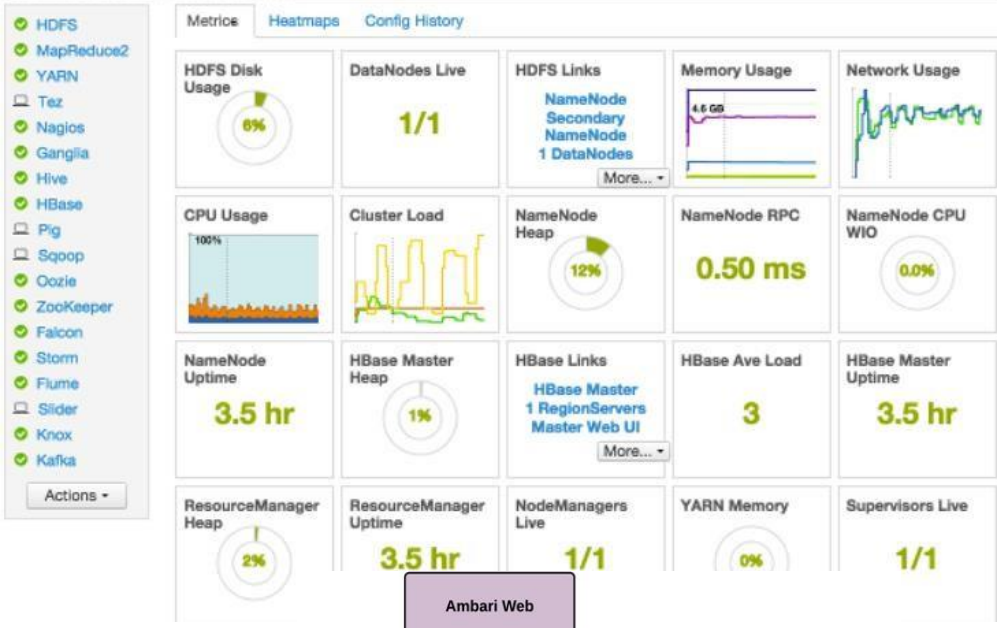
1. **Centralized Configuration Management**: ZooKeeper stores configuration data in a centralized manner, allowing distributed applications to access and modify configuration settings easily.
2. **Naming Service**: It provides a naming registry for distributed services, allowing them to find each other and manage resources dynamically.
3. **Synchronization**: ZooKeeper helps synchronize distributed processes, ensuring they work in coordination and without conflicts.
4. **Group Management**: Supports the creation and management of groups, enabling services to join and leave groups dynamically.
5. **Watchers**: ZooKeeper can set watchers on nodes to get notified of changes, allowing applications to react to changes in state.

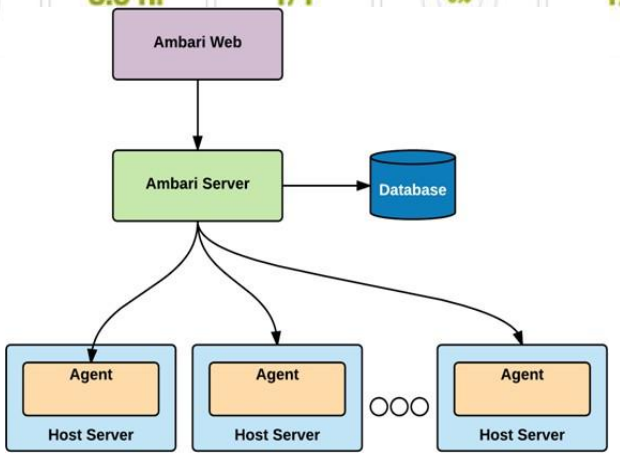


Apache Ambari

Apache Ambari

- Apache Ambari is a web-based tool designed for provisioning, managing, and monitoring Apache Hadoop clusters.
- It provides a user-friendly interface that simplifies the management of complex Hadoop environments, making it easier for administrators and operators to deploy and manage various components of the Hadoop ecosystem.
- **Key Features of Apache Ambari**
  - **Cluster Provisioning**: Ambari simplifies the installation and configuration of Hadoop clusters, allowing users to easily add or remove nodes and services through a guided installation wizard.
  - **Service Management**: Users can start, stop, and restart various Hadoop services (such as HDFS, YARN, Hive, HBase, etc.) through the Ambari interface.
  - **Monitoring and Alerts**: Ambari provides dashboards for real-time monitoring of cluster health, service metrics, and resource usage. It also supports setting up alerts for various cluster events or metrics.
  - **Configuration Management**: It allows for easy management of configuration settings for Hadoop services, including the ability to apply configurations in a centralized manner.
  - **User Management**: Administrators can manage users and their permissions, ensuring secure access to cluster resources and configurations.
  - **Integration with Ambari Metrics and Alerts**: Supports the collection and visualization of metrics from different Hadoop components, allowing for comprehensive monitoring and alerting.





Hadoop Components Summary

Other Hadoop Components

Category / Component	Description
Storage Components	
HDFS	Distributed file system for scalable and fault-tolerant data storage.
HBase	NoSQL database for real-time read/write access to large datasets.
Phoenix	SQL layer on top of HBase for low-latency OLTP and analytics.
Batch Processing	
MapReduce	Traditional batch processing model for parallel computation on large data.
Tez	DAG-based execution engine; faster alternative to MapReduce.
Hive	SQL-like interface for querying large datasets, often runs on Tez or Spark.
Pig	Dataflow scripting platform using Pig Latin for batch data processing.
Real-Time / Stream Processing	
Spark Streaming	Micro-batch stream processing engine within Apache Spark.
Flink	True stream processing engine with high throughput and low latency.
Storm	Distributed real-time computation system for event-driven processing.
Data Ingestion & Integration	
Sqoop	Imports/exports structured data between Hadoop and relational databases.
Flume	Collects and transports large volumes of log or event data.

Other Hadoop Components

Category / Component	Description
Security & Governance	
Ranger	Centralized security for authorization, policies, and auditing.
Knox	Gateway that provides perimeter security and REST API access to Hadoop.
Atlas	Metadata management and data governance framework.
Machine Learning & Search	
Mahout	Scalable machine learning library for classification, clustering, etc.
Solr	Enterprise search platform; supports full-text search on Hadoop data.
Cluster Coordination	
YARN	Resource management layer for job scheduling and cluster resource utilization.
Zookeeper	Coordination service for configuration, synchronization, and leader election.
Serialization & File Formats	
Avro	Row-based data serialization framework with schema support.
Parquet	Columnar storage format optimized for performance and compression.
ORC	Columnar storage format optimized for Hive; enables efficient querying.
UI, Workflow & Dev Tools	
Oozie	Workflow scheduler to orchestrate complex Hadoop jobs.
Ambari	Web UI for provisioning, monitoring, and managing Hadoop clusters.
Hue	Web-based user interface for querying and interacting with Hadoop tools.
Zeppelin	Interactive data analytics notebook that supports Spark, SQL, and more.